

SC402 – Introduction to Cryptography (3-0-0-3)

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Course Outlines

Cryptography is concerned with the mathematical, algorithmic, and implementation aspects of information and network security. The topics in this course include:

- **Classical cryptography vs. Modern cryptography** - affine cipher, vigenere cipher; Security definition and adversarial assumptions
- **Secret Key Cryptography (SKC)**: SPN, pseudo-random functions; Modular Arithmetic-1; Block ciphers; Stream ciphers; Modes of operation; Security notion – perfect security, semantic security, provable security, indistinguishability based security, CPA, CCA.
- **Hash functions, Pseudo-Random Generators** – Properties, Constructions; Message Authentication Codes; Security notion of MAC; Authenticated Encryptions.
- **Public Key Cryptography (PKC)**: Modular Arithmetic-2, RSA, DSA, El Gamal scheme; PKI, Key Management; Security notion – IND-CPA, IND-CKA.
- **Advanced Topics and Applications** - Secret sharing scheme; Secure Multiparty Computation; Identity-based cryptography.

Reading Materials

- **Introduction to Modern Cryptography** - Jonathan Katz and Yehuda Lindell, CRC Press.
- Cryptography: Theory and Practice - Douglas Stinson, Third edition, CRC Press.
- Handbook of Applied Cryptography - Alfred Menezes, Paul Oorschot and Scott Vanstone. Available [Online](#).

Grading Policy:

- Assignments, Class participation – 20%
- In-semester examination (one) – 30%
- End semester examination – 50%

Distribution of Lectures

Sl. No.	Description		No. of Lectures
1	Classical cryptography vs. Modern cryptography		5
	1.1	Classical cryptography – shift cipher, affine cipher, Vigenere cipher	2
	1.2	Security definition and adversarial assumptions. Security notion – perfect security, semantic security.	3

2	Symmetric Key Cryptography – SKC		11
	2.1	Building blocks of SKC – SPN, pseudo-random functions	2
	2.2	Block ciphers – Modular Arithmetic-1, AES, 3DES	3
	2.3	Stream ciphers – LFSR, RC4	2
	2.4	Modes of Operation – CBC, CFB, OFB, CTR	2
	2.5	Security notion – provable security, indistinguishability based security, CPA, CCA	2
3	Hash functions, Pseudo-Random Generators		6
	3.1	Hash functions – Properties, Constructions, SHA	3
	3.2	Message Authentication Codes (MAC): Security notions, CMA	2
	3.3	Authenticated Encryptions	1
4	Public Key Cryptography – PKC		11
	4.1	Modular Arithmetic-2, RSA, DSA, El Gamal scheme	6
	4.2	PKI, Key management, Digital Certificate	2
	4.3	Security notion – IND-CPA, IND-CKA	3
5	Advanced Topics and Applications		9
	5.1	Secret sharing scheme and applications.	2
	5.2	Secure Multiparty Computation - Oblivious transfer, Bit commitment, Zero-knowledge proof	3
	5.3	Introduction to ECC, Identity-based cryptography	4